Plane local testing of multiplicity codes via algebraic plane characterization

Dan Karliner, Roie Salama, Amnon Ta-Shma

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Editorial scope. Counted original Theorem64 (restating Theorem9), the quantitative plane tester for multiplicity codes. Complete Theorem60 plane-characterization argument and all local supporting algebraic and sampling proofs are bundled, not separately counted. Full author RM-to-MRM transfer proof is necessary context. Field-characteristic, degree and multiplicity constraints remain exactly as published. No author theorem, notation or proof is mathematically corrected.

Allowed standard prerequisites (editorial).

- Classical Grobner basis theory, Combinatorial Nullstellensatz with multiplicity [3], Lucas theorem, and imported Reed-Muller local tester [12]. Exact author imported statements, definitions and their source-local explanations are retained. All local representing-polynomial, coefficient-isolation, degree and plane-sampling proofs retained; no recursive reconstruction of cited background results.

2 Preliminaries (author context excerpts)

2.1 Notation

We denote vectors by bold letters. For $\boldsymbol{X} = X_1, \ldots, X_m$ we denote by $\mathbb{F}[\boldsymbol{X}]$ the set of multivariate polynomials in the variables $X_1, \ldots, X_m$. We denote by $\mathbb{F}[\boldsymbol{X}]^{\leq d}$ the set of polynomials of individual degree at most d, and by $\mathbb{F}[\boldsymbol{X}]^{\mathrm{loc}\leq d}$ the set of polynomials of individual degree at most d (i.e degree in each variable). Given a vector $\boldsymbol{I} = (i_1, \ldots, i_m) \in \mathbb{N}^m$, we use the notation

$$\boldsymbol{X}^{\boldsymbol{I}} \stackrel{\mathrm{def}}{=} \prod_{k=1}^{m} X_k^{i_k}.$$

2.2 Reed Muller and Multiplicity codes

Definition 10. Let d, m be non-negative integers, and q a prime power. The (m, d, q) Reed-Muller code is defined as the set of evaluation vectors of m-variate polynomials of degree $\leq d$ over $\mathbb{F}_q^m$, namely,

$$\mathrm{RM}(q, m, d) = \{(f(\alpha))_{\alpha \in \mathbb{F}_q^m} \mid f \in \mathbb{F}_q[\boldsymbol{X}]^{\leq d}\}.$$

We will make use of the following lemma:

Lemma 11 ([10], Lemma 3.2). Let $\delta^{\mathrm{RM}}_{q,m,d}$ be the relative distance of the code $\mathrm{RM}(q,m,d)$. Then $\delta^{\mathrm{RM}}_{q,m,d} \geq q^{-d/(q-1)}$.

Definition 12 (Hasse derivative). For a multivariate $P(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ where $\boldsymbol{X} = (X_1, \ldots, X_m)$ for some $m \in \mathbb{N}$, and a non-negative vector $\boldsymbol{I} \in \mathbb{N}^m$, the $\boldsymbol{I}$-th Hasse derivative of P, denoted $P^{(\boldsymbol{I})}(\boldsymbol{X})$, is the coefficient of $\boldsymbol{Z}^{\boldsymbol{I}}$ in the polynomial $P(\boldsymbol{X}, \boldsymbol{Z}) = P(\boldsymbol{X} + \boldsymbol{Z})$. Thus

$$P(\boldsymbol{X} + \boldsymbol{Z}) = \sum_{\boldsymbol{I}} P^{(\boldsymbol{I})}(\boldsymbol{X}) \cdot \boldsymbol{Z}^{\boldsymbol{i}}.$$

Hasse derivatives are linear. I.e, for all $P, Q \in \mathbb{F}[\boldsymbol{X}]$ and $\lambda \in \mathbb{F}$, $(\lambda P)^{(\boldsymbol{I})}(\boldsymbol{X}) = \lambda P(\boldsymbol{X})$ and $P^{(\boldsymbol{I})}(\boldsymbol{X}) + Q^{(\boldsymbol{I})}(\boldsymbol{X}) = (P+Q)^{(\boldsymbol{I})}(\boldsymbol{X})$. The product rule shows $(PQ)^{(\boldsymbol{I})}(\boldsymbol{X}) = \sum_{\boldsymbol{I}_0 + \boldsymbol{I}_1 = \boldsymbol{I}} P^{(\boldsymbol{I}_0)}(\boldsymbol{X}) \cdot Q^{\boldsymbol{I}_1}(\boldsymbol{X})$.

Definition 13 (Weight). If $\boldsymbol{I} = (i_1, \ldots, i_m) \in \mathbb{N}^m$ then $w(\boldsymbol{I}) = \sum_{j=1}^m i_j$.

Definition 14 (Multiplicity). For $P(\boldsymbol{X}) \in \mathbb{F}[\boldsymbol{X}]$ and $\boldsymbol{a} \in \mathbb{F}^m$, the multiplicity of P at $\boldsymbol{a}$, denoted $\mathrm{mult}(P, \boldsymbol{a})$, is the largest integer s such that for every non-negative vector $\boldsymbol{I}$ with $w(\boldsymbol{I}) < s$ we have $P^{(\boldsymbol{I})}(\boldsymbol{a}) = 0$. If s may be taken arbitrarily large, we set $\mathrm{mult}(P, \boldsymbol{a}) = \infty$. Note that by definition $\mathrm{mult}(P, \boldsymbol{a}) \geq 0$ for every $\boldsymbol{a}$. One important property about multiplicities is a generalization of the Schwartz-Zippel lemma for multivariate polynomials:

Lemma 15 ([6]). Let $P \in \mathbb{F}[\boldsymbol{X}]$ be a non-zero polynomial of total degree at most d. Then for any finite $A \subseteq \mathbb{F}$,

$$\sum_{\boldsymbol{a} \in A^m} \mathrm{mult}(P, \boldsymbol{a}) \leq d \cdot |A|^{m-1}.$$

Definition 16 (Multiplicity code). Let $m, d \geq s$ be non-negative integers, and let q be a prime power. Let

$$\Sigma_{m,s} = \mathbb{F}_q^{\{\boldsymbol{I} : w(\boldsymbol{I}) < s\}} \simeq \mathbb{F}_q^{\binom{m+s-1}{m}}.$$

For $P(\boldsymbol{X}) \in \mathbb{F}_q[X_1, \ldots, X_m]$ we define the order s evaluation of P at $\boldsymbol{a}$, denoted $P^{(<s)}(\boldsymbol{a})$, to be the vector $(P^{(\boldsymbol{I})}(\boldsymbol{a}))_{\boldsymbol{I} : w(\boldsymbol{I}) < s} \in \Sigma_{m,s}$. The multiplicity code $\mathrm{MRM}(q, m, d, s)$ is defined as follows. The alphabet of the code is $\Sigma_{m,s}$ and the length is q^m. Every polynomial $P(\boldsymbol{X}) \in \mathbb{F}_q[\boldsymbol{X}]$ of $\deg(P) \leq d$ defines a codeword by $(P^{(<s)}(\boldsymbol{a}))_{\boldsymbol{a} : \boldsymbol{a} \in \mathbb{F}_q^m} \in (\Sigma_{m,s})^{q^m}$. We also let $\mathrm{MRS}(q, d, s) := \mathrm{Mult}(q, 1, d, s)$ stand for Reed-Solomon multiplicity code. The following lemma states the relationship between the derivatives of a polynomial to the derivatives of its restriction to a line.

Lemma 17 ([15], Sec 4). Let $P \in \mathbb{F}[\boldsymbol{X}]$ be a multivariate polynomial where $\boldsymbol{X} = (X_1, \ldots, X_m)$. Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{F}^m$ and define a univariate polynomial $P|_{L_{\boldsymbol{a},\boldsymbol{b}}}(t) = P(\boldsymbol{a}t + \boldsymbol{b})$. Then

$$(P|_{L_{\boldsymbol{a},\boldsymbol{b}}})^{(j)}(t) = \sum_{\boldsymbol{I} : w(\boldsymbol{I}) = j} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}) \cdot \boldsymbol{a}^{\boldsymbol{I}}.$$

3 Polynomials and tables (author context excerpts)

A table is an element of $\Sigma_{m,s}^{q^m}$, i.e., a function mapping every evaluation point in $\mathbb{F}_q^m$ to an element in $\Sigma_{m,s}$. EVAL takes a multi-variate polynomial and returns its table of evaluations. More precisely,

Definition 39. We define $\mathrm{EVAL}_{m,s} : \mathbb{F}_q[\boldsymbol{X}] \times \mathbb{F}_q^m \to \Sigma_{m,s}$ by

$$\mathrm{EVAL}_{m,s}(P; \boldsymbol{a}) = (P^{(\boldsymbol{I})}(\boldsymbol{a}))_{w(\boldsymbol{I})<s}.$$

Similarly, we define $\mathrm{EVAL}_{m,s} : \mathbb{F}_q[\boldsymbol{X}] \to (\Sigma_{m,s})^{q^m}$ by

$$\mathrm{EVAL}_{m,s}(P) = (\mathrm{EVAL}(P; \boldsymbol{a}))_{\boldsymbol{a}\in\mathbb{F}_q^m}.$$

We say $T \in (\Sigma_{m,s})^{q^m}$ is the table of $P \in \mathbb{F}_q[\boldsymbol{X}]$ if $\mathrm{EVAL}(P) = T$. An element in $\Sigma_{m,s}$ contains information about all the derivatives of order up to s. Sometimes we would like to focus on a certain directional derivative. If $\sigma \in \Sigma_{m,s}$ and $\boldsymbol{I} \in \mathbb{N}^m$ with $w(\boldsymbol{I}) < s$, then $\sigma_{(\boldsymbol{I})} \in \mathbb{F}_q$ is the entry of σ that encodes the $\boldsymbol{I}$'th derivative. Similarly, if $f \in \Sigma_{m,s}^{q^m}$ is a table, i.e., $f : \mathbb{F}_q^m \to \Sigma_{m,s}$, then $f_{(\boldsymbol{I})} : \mathbb{F}_q^m \to \mathbb{F}_q$ is defined by letting $f_{(\boldsymbol{I})}(\boldsymbol{x})$ be the $\boldsymbol{I}$'th entry of $f(\boldsymbol{x}) \in \Sigma_{m,s}$. To understand the low-dimensional tests on tables, we need to define the restriction of tables to subspaces. If T is a table $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ and we want to restrict to the plane $\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}$ the restriction $T_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}$ should be a table $\mathbb{F}_q^2 \to \Sigma_{2,s}$. To this end we define the alphabet reduction map $\phi_{(\boldsymbol{a},\boldsymbol{b})} : \Sigma_{m,s} \to \Sigma_{2,s}$ by

Definition 48.

$$(\phi_{(\boldsymbol{a},\boldsymbol{b})})(\boldsymbol{z})_{\boldsymbol{J}=(j_1,j_2)} = \sum_{\boldsymbol{I}\in\mathbb{N}^m} z_{\boldsymbol{I}} \cdot \sum_{\substack{\boldsymbol{I}_1+\boldsymbol{I}_2=\boldsymbol{I} \\ w(\boldsymbol{I}_1)=j_1, w(\boldsymbol{I}_2)=j_2}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1} \boldsymbol{a}^{\boldsymbol{I}_1}\boldsymbol{b}^{\boldsymbol{I}_2}.$$

This map applies the "chain rule" to an element in $\Sigma_{m,s}$, in accordance with Lemma 18. We may then define

$$T_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} = \phi_{(\boldsymbol{a},\boldsymbol{b})} \circ T \circ \ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}.$$

Similarly, for $\boldsymbol{h} = (\boldsymbol{h}_0, \boldsymbol{h}_1, \ldots, \boldsymbol{h}_k)$ we define $\phi_{\boldsymbol{h}}$ and $T_{\boldsymbol{h}}$ by:

$$(\phi_{\boldsymbol{h}}(\boldsymbol{z}))_{\boldsymbol{J}} = \sum_{\boldsymbol{I}\in\mathbb{N}^m} z_{\boldsymbol{I}} \cdot \sum_{\substack{\boldsymbol{I}_1+\cdots+\boldsymbol{I}_k=\boldsymbol{I} \\ w(\boldsymbol{I}_r)=j_r}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1, \ldots, \boldsymbol{I}_k} \prod_{i=1}^{k} \boldsymbol{h}_k^{\boldsymbol{I}_k}, \text{ and,}$$

$$T_{\boldsymbol{h}}(\boldsymbol{Y}) = \phi_{(\boldsymbol{h})} \circ T\left(\boldsymbol{h}_0 + \sum_{i=1}^{k} Y_i \boldsymbol{h}_i\right).$$

5 Local testing above the RM testing dimension

In this section we look at the local characterization and local testing of $\mathrm{MRM}(q, m, d, s)$ by dimension k tests, when k is above the RM testing dimension $t_{q,d} = \lceil (d+1)/(q - q/p) \rceil$ of $\mathrm{RM}(q, m, d)$. By Theorem 2 dimension k subspaces give a local

test (and hence also a local characterization) for $\mathrm{RM}(q,m,d)$. We show they also give a local test (and hence also a local characterization) for $\mathrm{MRM}(q,m,d,s)$ for $s<q$. There is, however, some parameter loss in the reduction as we next explain. To formally state the result we need some notation. For $\boldsymbol{x}\in\mathbb{F}_q^m$ let

$$H_k=\{\boldsymbol{h}=(\boldsymbol{h}_0,\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k)\mid \boldsymbol{h}_0,\ldots,\boldsymbol{h}_k\in\mathbb{F}_q^m,\dim(\mathrm{span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\})=k\},\text{ and,}$$
$$H_{k,\boldsymbol{x}}=\{\boldsymbol{h}\in H_k\mid \boldsymbol{x}\in\boldsymbol{h}_0+\mathrm{Span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\}\}.$$

By $\boldsymbol{h}\sim_k H$ (resp. $H_{k,\boldsymbol{x}}$) we mean a choosing $\boldsymbol{h}$ uniformly at random from H_k (resp. $H_{k,\boldsymbol{x}}$). We let $A_{\boldsymbol{h}}=\boldsymbol{h}_0+\mathrm{Span}\{\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k\}$ be the k-dimensional affine space defined by $\boldsymbol{h}$. We also recall the distance and rejection function from the introduction:

Definition 50. Let $T:\mathbb{F}_q^m\to\Sigma_{m,s}$ be a table. Then

- $\delta(T,\mathrm{MRM}(q,m,d,s))$ is the distance between T and the closest evaluation table of a degree d polynomial, i.e., $\delta(T,\mathrm{MRM}(q,m,d,s))=\min_{G\in\mathrm{MRM}(q,m,d,s)}\{\delta(T,G)\}$, and,
- $\mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)$ is the probability a dimension-k test demonstrates that T is not a degree d polynomial.

With this notation we prove:

Theorem 51. Let $\mathbb{F}_q$ be a field of size q, and assume $s\le\min\{d,q-1\}$. Suppose for $\mathrm{RM}(q,m,d)$ there exists $\alpha>0$ and $c_0\le 1$ such that for every f

$$\mathrm{REJ}_{k,d}^{\mathrm{RM}}(f)\ge\min\{\alpha\cdot\delta(f,\mathrm{RM}(q,m,d)),c_0\}.$$

Then, for every $s<q$, for every T we have

$$\mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)\ge\min\{\alpha'\cdot\delta(T,\mathrm{MRM}(q,m,d,s)),c_0\} \tag{6}$$

for

$$\alpha'=\left(1-\frac{s-1}{q}\right)\frac{1}{1+q^{d/(q-1)}\frac{1}{\alpha}}. \tag{7}$$

For example, assume q is prime. If $d+1\le q-1$ the testing dimension is 1, meaning that lines are a good local test for $\mathrm{RM}(q,m,d)$. Then, Theorem 51 says that lines are also a good test for $\mathrm{MRM}(q,m,d,s)$, alas, with a larger coefficient. Similarly, if $q<d\le 2(q-1)$ the RM testing dimension is 2 and therefore planes are a good local test for $\mathrm{RM}(q,m,d)$. Then, Theorem 51 says that planes are also a good test for $\mathrm{MRM}(q,m,d,s)$, with a larger coefficient. In the same vein 3-dimensional planes are a good local test for $\mathrm{MRM}(q,m,d,s)$ when $d\le 3(q-1)$. To explain the intuition behind the theorem we first consider the characterization aspect of it. Suppose k is above the RM testing dimension, and that the table T passes all $\mathrm{MRM}(q,m,d,s)$ k-dimensional tests $\boldsymbol{h}$. Specifically, this means that for every $\boldsymbol{h}\in H_k$, the table T restricted to the k-dimensional affine space $A_{\boldsymbol{h}}$ is consistent with a degree d polynomial $P_{\boldsymbol{h}}$. Now, let $T_{(0)}$ be the table T where at each entry we keep only the evaluation of the function itself and remove the evaluations that are associated with higher derivatives. Then, in particular, for every $\boldsymbol{h}\in H_k$, the table $T_{(0)}$ restricted to the k-dimensional

space $A_{\boldsymbol{h}}$ is still consistent with the degree d polynomial $P_{\boldsymbol{h}}$. As k is above the RM testing dimension, there must be a unique degree d polynomial P that is consistent with $P_{\boldsymbol{h}}$ (and the table $T_{(0)}$) for every $\boldsymbol{h} \in H_k$. This P is the only possible candidate for a low-degree explanation of the table T. What we need to check, and is indeed correct, is that since T passes all dimension k tests, P is indeed consistent with T. The testing case requires more technical details but is similar in spirit. We first, again, look at $T_{(0)}$ that contains only the function evaluations, and not the higher derivatives evaluations. If T passes the test with high enough probability, then so does $T_{(0)}$, and this ensures the existence of a global degree-d polynomial P that agrees with most values of $T_{(0)}$. Again, what remains to be shown is that P agrees with most values of T, which we indeed prove. In short, one can informally say that Theorem 51 shows that the MRM testing dimension is not larger than the RM testing dimension, and that above the RM testing dimension one can get local testing for MRM codes. A natural question is whether the MRM testing dimension is equal to the RM testing dimension, or not. In Section 4 we saw that when the RM testing dimension is larger than 1, so is the MRM testing dimension, and lines do not characterize the MRM code. One might be drawn to the conjecture that the RM and MRM testing dimensions coincide. However, surprisingly, in Section 6 we show that no matter what the RM testing dimension is, when $d < sq$ the MRM testing dimension is at most two (for a precise statement see Theorem 60). For example, for $\mathrm{MRM}(q, d, m, s)$ with $2q < d < 3(q-1)$, the RM testing dimension is (roughly) 3 while the MRM testing dimension is still 2.

5.1 The proof

Proof. Let $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ be a table. Let

$$\rho = \mathrm{REJ}_{k,d}^{\mathrm{MRM}}(T)$$

be the k-dimensional MRM test rejection probability. If $\rho \geq c_0$ we are done. Therefore, we assume $\rho < c_0$. We first utilize what we know at the zero level.

Claim 53. Let $\delta^{\mathrm{RM}} = \delta_{q,m,d}^{\mathrm{RM}}$ be the distance of the $\mathrm{RM}(q, m, d)$ code. There exists a degree d polynomial P such that

$$\Pr_{\boldsymbol{h} \in H_k}\left((\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} \neq \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})\right) \stackrel{\mathrm{def}}{=} \varepsilon_0 \leq \rho\left(1 + \frac{1}{\alpha \cdot \delta^{\mathrm{RM}}}\right).$$

We call $\boldsymbol{h}$ good if $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$ and bad otherwise. In this terminology, $\varepsilon_0 = \Pr_{\boldsymbol{h} \in H_k}[\boldsymbol{h} \text{ is bad}]$.

Proof. Let $T_{(0)} : \mathbb{F}_q^m \to \mathbb{F}_q$ be as in Section 3, i.e., the table where we keep only the entries of the function evaluations and ignore the evaluations of higher order derivatives. For a given affine space A we have that $(T|_A)_{(0)} = T_{(0)}|_A$, and so if the former agrees with a degree d polynomial, so does the latter. It follows that

$$\mathrm{REJ}_{k,d}^{\mathrm{RM}}(T_{(0)}) \leq \rho.$$

By the hypothesis regarding $\mathrm{RM}(q, m, d)$ and using $\rho < c_0$ there exists a unique degree d polynomial $P \in \mathbb{F}_q[\boldsymbol{X}]$ such that

$$\delta_{(0)} \stackrel{\mathrm{def}}{=} \delta(T_{(0)}, P) \leq \rho/\alpha.$$

We notice that

$$\begin{aligned}\mathbb{E}_{\boldsymbol{h}\sim H_k}\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) &= \Pr_{\boldsymbol{x}\sim\mathbb{F}_q^m}(T_{(0)}(\boldsymbol{x}) \neq P(\boldsymbol{x})) \\ &= \delta(T_{(0)}, P) = \delta_{(0)},\end{aligned}$$

because the subspaces in $A_{\boldsymbol{h}}$ for $\boldsymbol{h} \in H_k$, cover every point an equal number of times. Therefore, by Markov's inequality,

$$\Pr_{\boldsymbol{h}\sim H_k}(\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) \geq \delta^{\mathrm{RM}}) \leq \frac{\delta_{(0)}}{\delta^{\mathrm{RM}}}. \tag{8}$$

Also, by assumption,

$$\Pr_{\boldsymbol{h}\in H_k}((\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} \notin \mathrm{MRM}(q,k,d,s)) \leq \rho. \tag{9}$$

This means that except for probability ρ over $\boldsymbol{h}$, $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}}$ agrees with $\mathrm{EVAL}(P_{\boldsymbol{h}})$ for some k-variate degree d polynomial $P_{\boldsymbol{h}}$. When this happens it also holds that $T_{(0)}$ agrees with $P_{\boldsymbol{h}}$ over $A_{\boldsymbol{h}}$ (because $\phi_{\boldsymbol{h}}$ does not change the zero level). Thus, Equations (8) and (9) together imply that except for probability $\rho + \delta_0/\delta^{\mathrm{RM}} \leq \rho(1 + 1/(\alpha\delta^{\mathrm{RM}}))$ over $\boldsymbol{h}$, we simultaneously have that $T_{(0)} = P_{\boldsymbol{h}}$ over $A_{\boldsymbol{h}}$ and that $\delta(T_{(0)}|_{A_{\boldsymbol{h}}}, P|_{A_{\boldsymbol{h}}}) < \delta^{\mathrm{RM}}$. When both events happen we conclude that

$$\delta(P_{\boldsymbol{h}}, P|_{A_{\boldsymbol{h}}}) < \delta^{\mathrm{RM}}.$$

As $P_{\boldsymbol{h}}$ and $P|_{A_{\boldsymbol{h}}}$ are degree d polynomials on $A_{\boldsymbol{h}}$ and are closer than δ^{RM}, it must be the that in fact $P_{\boldsymbol{h}} = P|_{A_{\boldsymbol{h}}}$. Thus, $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}}$ is a valid $\mathrm{MRM}(q,k,d,s)$ table, and it is the table of the polynomial $P|_{A_{\boldsymbol{h}}}$, i.e., $(\phi_{\boldsymbol{h}} \circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$ as desired. $\triangleleft$ Our goal is to bound $\delta(\mathrm{EVAL}(P), T)$. This means that at most $\boldsymbol{x} \in \mathbb{F}_q^m$ we should have $T(\boldsymbol{x}) = \mathrm{EVAL}(P; \boldsymbol{x})$. $T(\boldsymbol{x})$ and $\mathrm{EVAL}(P; \boldsymbol{x})$ contains values for all the m-variate directional derivatives of order up to s. Our handle on these values is Claim 53, that shows that most $\boldsymbol{h}$ are good, meaning that the k-variate derivatives of $P|_{A_{\boldsymbol{h}}}$ and $\phi_{\boldsymbol{h}} \circ T|_{A_{\boldsymbol{h}}}$ are the same. We notice that every such k-variate derivative is a linear combination (dependent on $\boldsymbol{h}$) of the m-variate derivatives. If $\boldsymbol{x}$ is such that for many $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$, $\boldsymbol{h}$ is good, then for that $\boldsymbol{x}$ we get many linear equations on the m-variate derivatives. Our task is to prove that for many $\boldsymbol{x}$ there are enough good $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$ to force the underlying m-variate derivatives of P and T to agree.

Claim 54. We say $\boldsymbol{x} \in \mathbb{F}_q^m$ is bad if $\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}[\boldsymbol{h} \text{ is bad}] \geq 1 - (s-1)/q$ and good otherwise. Then

$$\Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}[\boldsymbol{x} \text{ is bad}] \leq \frac{q}{q-(s-1)} \cdot \Pr_{\boldsymbol{h}\in H_k}[\boldsymbol{h} \text{ is bad}].$$

Proof. We have

$$\mathbb{E}_{\boldsymbol{x}\in\mathbb{F}_q^m}\left[\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad})\right] = \Pr_{\boldsymbol{h}\in H_k}(\boldsymbol{h} \text{ is bad}) = \varepsilon_0,$$

where ε_0 is as in Claim 53. This is because choosing a uniform $\boldsymbol{h} \in H_k$ is the same as first choosing a uniform $\boldsymbol{x} \in \mathbb{F}_q^m$ and then choosing a uniform $\boldsymbol{h} \in H_{k,\boldsymbol{x}}$.

Therefore, for every $c > 1$, by Markov's inequality,

$$\Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}\left[\Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad}) \geq c\cdot\varepsilon_0\right] \leq 1/c.$$

Choosing $c = (q-(s-1))/(q\cdot\varepsilon_0)$ gives the result. ◁ We note that if most k-dimensional subspaces are good, then most lines are:

Claim 55. For every $\boldsymbol{x}\in\mathbb{F}_q^m$ and $k \geq 1$,

$$\Pr_{\boldsymbol{u}\in H_{1,\boldsymbol{x}}}[\boldsymbol{u} \text{ is bad}] \leq \Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}[\boldsymbol{h} \text{ is bad}].$$

Proof. Fix $\boldsymbol{x}$. For every $\boldsymbol{h}\in H_{k,\boldsymbol{x}}$, If $\boldsymbol{h}$ is good, then $(\phi_{\boldsymbol{h}}\circ T)|_{A_{\boldsymbol{h}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{h}}})$. This implies, in particular, that for every $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ such that $A_{\boldsymbol{u}}\subseteq A_{\boldsymbol{h}}$, we also have that $(\phi_{\boldsymbol{u}}\circ T)|_{A_{\boldsymbol{u}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}})$. The result then follows because we can sample $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ by first sampling $\boldsymbol{h}\in A_{k,\boldsymbol{x}}$ and then choosing a random $\boldsymbol{u}\in H_{1,\boldsymbol{x}}$ such that $A_{\boldsymbol{u}}\subseteq A_{\boldsymbol{h}}$. ◁ We now fix any good $\boldsymbol{x}$. We will prove:

Claim 56. Suppose $\boldsymbol{x}\in\mathbb{F}_q^m$ is good. Then for any m-variate direction $\boldsymbol{I}$ with $w(\boldsymbol{I}) < s$ we have

$$T_{(\boldsymbol{I})}(\boldsymbol{x}) = P^{(\boldsymbol{I})}(\boldsymbol{x}). \tag{10}$$

Once we prove the claim we can conclude the proof of the theorem because:

$$\begin{aligned}\delta(T,\mathrm{EVAL}(P)) &\leq \Pr_{\boldsymbol{x}\in\mathbb{F}_q^m}(\boldsymbol{x} \text{ is bad}) \leq \frac{q}{q-(s-1)}\cdot\varepsilon_0\\ &\leq \frac{q}{q-(s-1)}\cdot\rho\cdot\left(1+\frac{1}{\alpha\cdot\delta^{\mathrm{RM}}}\right)\\ &\leq \rho\left(1-\frac{q}{s-1}\right)^{-1}\left(1+\frac{q^{d/(q-1)}}{\alpha}\right).\end{aligned}$$

Proof of Claim 56. Fix a good $\boldsymbol{x}\in\mathbb{F}_q^m$ and let $w_0 < s$. We will show Equation (10) simultaneously for all m-variate directions $\boldsymbol{I}$ with $w(\boldsymbol{I}) = w_0$. We know that

$$\Pr_{\boldsymbol{u}\in H_{1,\boldsymbol{x}}}(\boldsymbol{u} \text{ is bad}) \leq \Pr_{\boldsymbol{h}\in H_{k,\boldsymbol{x}}}(\boldsymbol{h} \text{ is bad}) < 1-\frac{s-1}{q}, \tag{11}$$

where the first inequality is by Claim 55 and the second because $\boldsymbol{x}$ is good. Suppose $\boldsymbol{u} = (\boldsymbol{b},\boldsymbol{a})$ is good (where $\boldsymbol{b},\boldsymbol{a}\in\mathbb{F}_q^m$). Then, by definition, $(\phi_{\boldsymbol{u}}\circ T)|_{A_{\boldsymbol{u}}} = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}})$. In particular $\phi_{\boldsymbol{u}}(T(\boldsymbol{x})) = \mathrm{EVAL}(P|_{A_{\boldsymbol{u}}};\boldsymbol{x})$, because $\boldsymbol{x}\in A_\ell$. Now:

- By Definition 48 (and plugging $k=1$)

$$(\phi_{\boldsymbol{u}}(T(\boldsymbol{x})))_{w_0} = \sum_{\boldsymbol{I},w(\boldsymbol{I})=w_0} T(\boldsymbol{x})_{(\boldsymbol{I})}\cdot\boldsymbol{a}^{\boldsymbol{I}},$$

- Also, by Lemma 17 (plugging $t=0$),

$$(\mathrm{EVAL}(P|_{A_{\boldsymbol{u}}};\boldsymbol{x}))_{w_0} = \sum_{\boldsymbol{I},w(\boldsymbol{I})=w_0} P^{\boldsymbol{I}}(\boldsymbol{x})\cdot\boldsymbol{a}^{\boldsymbol{I}}.$$

Thus, a good $\boldsymbol{u} = (\boldsymbol{b}, \boldsymbol{a})$ gives the liner equation

$$\sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (P^{\boldsymbol{I}}(\boldsymbol{x}) - T(\boldsymbol{x})_{(\boldsymbol{I})}) \cdot \boldsymbol{a}^{\boldsymbol{I}} = 0,$$

where the variables are $v_{\boldsymbol{I}} = P^{\boldsymbol{I}}(\boldsymbol{x}) - T(\boldsymbol{x})_{(\boldsymbol{I})}$ for every m-variate direction $\boldsymbol{I}$ of total weight exactly w_0. Thus, Equation (11) implies that

$$\Pr_{\boldsymbol{a} \in \mathbb{F}_q^m \setminus \{\boldsymbol{0}\}} \left(\sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})) \cdot \boldsymbol{a}^{\boldsymbol{I}} = 0 \right) > \frac{s-1}{q}.$$

Now, look at the polynomial $f_{\boldsymbol{x}} \in \mathbb{F}_q[X_1, \ldots, X_m]$ defined by

$$f_{\boldsymbol{x}}(\boldsymbol{a}) = \sum_{\boldsymbol{I}, w(\boldsymbol{I})=w_0} (T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})) \cdot \boldsymbol{a}^{\boldsymbol{I}}.$$

$f_{\boldsymbol{x}}$ is an m-variate, degree w_0 homogeneous polynomial, and it is 0 with probability larger than $(s-1)/q \geq w_0/q = \deg(f_{\boldsymbol{x}})/q$. By the Schwartz-Zippel lemma, it must be the 0 polynomial. Therefore, $T(\boldsymbol{x})_{(\boldsymbol{I})} - P^{(\boldsymbol{I})}(\boldsymbol{x})$ for all $\boldsymbol{I}$ with $w(\boldsymbol{I}) = w_0$ as desired. ◁

◁

Remark 57. Another way to view the argument, is that each $\boldsymbol{a} \in \mathbb{F}_q^m$ is an evaluation point of a homogeneous $\mathrm{RM}(q, m, w_0)$ codeword, and therefore each good $\boldsymbol{u} = (\boldsymbol{b}, \boldsymbol{a})$ gives a zero coordinate of the codeword. If the number of good $\boldsymbol{u}$ is too large, we get too many zeroes, and therefore the codeword must be the zero codeword, meaning that the values of the variables are zero as we wish.

Remark 58. The argument we use has the information that many k-dimensional restrictions are good, but then chooses to reduce this knowledge to the weaker statement that for many $\boldsymbol{x}$, for many lines passing through $\boldsymbol{x}$, the linear restrictions are good. It seems that using the stronger statement might give a better code and improve the parameters, but we have not succeeded yet in analyzing this.

Additional author algebraic context

Recall The definition of the binomial and multinomial coefficients for natural numbers:

$$\binom{a}{b} \stackrel{\text{def}}{=} \frac{a!}{b!(a-b)!}, \qquad \binom{a}{b_1, \ldots, b_\ell} = \frac{a!}{b_1! \cdots b_\ell!},$$

where $\sum b_i = a$. We extend this definition to $\boldsymbol{I}, \boldsymbol{I}_1, \boldsymbol{J}, \boldsymbol{J}_1, \ldots, \boldsymbol{J}_\ell \in \mathbb{N}^m$ where $\boldsymbol{J} = \boldsymbol{J}_1 + \ldots + \boldsymbol{J}_\ell$ by

$$\binom{\boldsymbol{I}}{\boldsymbol{I}_1} \stackrel{\text{def}}{=} \prod_{k=1}^{m} \binom{I_k}{(I_1)_k}, \qquad \binom{\boldsymbol{J}}{\boldsymbol{J}_1, \ldots, \boldsymbol{J}_\ell} \stackrel{\text{def}}{=} \prod_{k=1}^{m} \binom{J_k}{(J_1)_k, \cdots, (J_\ell)_k}.$$

We mention Lucas theorem, that if p is prime, $q = p^w$ for $w \in \mathbb{N}$, and $a, b \in \mathbb{N}$ have base p representation $a = \sum_{\ell=0}^{w} a_\ell p^\ell$, $b = \sum_{\ell=0}^{w} b_\ell p^\ell$ with $0 \le a_\ell, b_\ell < p$, then

$$\binom{a}{b} \bmod p = \prod_{\ell=0}^{w} \binom{a_\ell}{b_\ell}, \tag{4}$$

where we use the convention that $\binom{c}{d} = 0$ when $d > c$. Thus, $\binom{a}{b} \bmod p \neq 0$ iff $a_\ell \ge b_\ell$ for all $\ell = 0, \ldots, w-1$. Finally, we let $g \in \mathbb{F}_q[X]$ denote the polynomial $g(X) = X^q - X$. For $\boldsymbol{X} = (X_1, \ldots, X_m)$ and $\boldsymbol{I} = (i_1, \ldots, i_m) \in \mathbb{N}^m$ we let $g(\boldsymbol{X})^{\boldsymbol{I}}$ denote $\prod_{k=1}^{m} (g(X_k))^{i_k}$.

Lemma 18. Let $P \in \mathbb{F}[\boldsymbol{X}]$ be a multivariate polynomial where $\boldsymbol{X} = (X_1, \ldots, X_m)$. Let $\boldsymbol{a}, \boldsymbol{b} \in \mathbb{F}^m$ and define a bivariate polynomial by $P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) = P(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c})$. Then for $(j_1, j_2) \in \mathbb{N}^2$:

$$(P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}})^{(j_1,j_2)}(t, r) = \sum_{\boldsymbol{I} \in \mathbb{N}^m} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}) \cdot \sum_{\substack{\boldsymbol{I}_1 + \boldsymbol{I}_2 = \boldsymbol{I} \\ w(\boldsymbol{I}_1) = j_1, w(\boldsymbol{I}_2) = j_2}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1} \boldsymbol{a}^{\boldsymbol{I}_1} \boldsymbol{b}^{\boldsymbol{I}_2}.$$

Proof. Given $R_1, R_2 \in \mathbb{F}$, we write the expression $P(\boldsymbol{a}(t+R_1) + \boldsymbol{b}(r+R_2) + \boldsymbol{c})$ in two different ways. On the one hand,

$$\begin{aligned} P(\boldsymbol{a}(t + R_1) + \boldsymbol{b}(r + R_2) + \boldsymbol{c}) &= P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t + R_1, r + R_2) \\ &= P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}((t, r) + (R_1, R_2)) \\ &= \sum_{j_1, j_2 \in \mathbb{N}} (P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}})^{(j_1,j_2)}(t, r) R_1^{j_1} R_2^{j_2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} P(\boldsymbol{a}(t + R_1) + \boldsymbol{b}(r + R_2) + \boldsymbol{c}) &= P(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c} + R_1\boldsymbol{a} + R_2\boldsymbol{b}) \\ &= \sum_{\boldsymbol{I} \in \mathbb{N}^m} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}) \cdot (R_1\boldsymbol{a} + R_2\boldsymbol{b})^{\boldsymbol{I}} \\ &= \sum_{\boldsymbol{I} \in \mathbb{N}^m} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}) \cdot \prod_{k=1}^{m} (a_k R_1 + b_k R_2)^{i_k} \\ &= \sum_{\boldsymbol{I} \in \mathbb{N}^m} P^{(\boldsymbol{I})}(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}) \sum_{\substack{\boldsymbol{I}_1 + \boldsymbol{I}_2 = \boldsymbol{I} \\ w(\boldsymbol{I}_1) = j_1 \\ w(\boldsymbol{I}_2) = j_2}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1} \boldsymbol{a}^{\boldsymbol{I}_1} \boldsymbol{b}^{\boldsymbol{I}_2} R_1^{j_1} R_2^{j_2}. \end{aligned}$$

Comparing coefficients of $R_1^{j_1} R_2^{j_2}$ for every $\boldsymbol{J} = (j_1, j_2) \in \mathbb{N}^2$ we get the result. ◁ We will also be interested in the restrictions of polynomials to general k-dimensional subspaces. Let $P|_{P_{\boldsymbol{h}_0,\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k}}(\boldsymbol{Y}) = P(\boldsymbol{h}_0 + \sum_{i=1}^{k} \boldsymbol{h}_i Y_i)$. Then, similarly to Lemma 18,

Lemma 19.

$$(P|_{P_{\boldsymbol{h}_0,\boldsymbol{h}_1,\ldots,\boldsymbol{h}_k}})^{\boldsymbol{J}}(\boldsymbol{Y}) = \sum_{\boldsymbol{I} \in \mathbb{N}^m} P^{(\boldsymbol{I})}\left(\boldsymbol{h}_0 + \sum_{i=1}^{k} \boldsymbol{h}_i Y_i\right) \cdot \sum_{\substack{\boldsymbol{I}_1 + \cdots + \boldsymbol{I}_k = \boldsymbol{I} \\ w(\boldsymbol{I}_r) = j_r}} \binom{\boldsymbol{I}}{\boldsymbol{I}_1, \ldots, \boldsymbol{I}_k} \prod_{i=1}^{k} \boldsymbol{h}_k^{\boldsymbol{I}_k}.$$

The proof is identical to the proof of Lemma 18.

2.3 Grobner bases and Nullstellensatz

The theory of Grobner bases describes the structure of ideals in the ring $R = \mathbb{F}[\boldsymbol{X}]$ and we briefly explain some of the essential concepts of this theory. We refer to [5] for a thorough treatment of this theory.

Definition 20. A monomial order $\succ$ on R is a relation $\succ$ on $\mathbb{Z}^n_{\geq 0}$, or equivalently a relation on the set of monomials x^α, $\alpha \in \mathbb{Z}^n_{\geq 0}$ satisfying:

1. $\succ$ is a total ordering.
2. If $\alpha \succ \beta$ and $\gamma \in \mathbb{Z}^n_{\geq 0}$ then $\alpha + \gamma \succ \beta + \gamma$.
3. $\succ$ is a well-ordering. I.e, every non-empty $A \subset \mathbb{Z}^n_{\geq 0}$ has a minimal element.

Example 21 (Lexicographic order). Let $\alpha, \beta \in \mathbb{Z}^m_{\geq 0}$. We say $\alpha \succ_{\text{lex}} \beta$ if the minimal i which satisfies $\alpha_i \neq \beta_i$, also satisfies $\alpha_i > \beta_i$.

Example 22 (Total degree lexicographic order). The total degree lexicographic order is defined as follows: A monomial m_1 is greater than m_2 if it has higher total degree, where ties are broken lexicographically (i.e $X_1 > X_2 > \cdots > X_m$). More formally, let $\alpha, \beta \in \mathbb{Z}^m_{\geq 0}$. Then $\alpha \succ_{\text{tot}} \beta$ if $w(\alpha) = \sum \alpha_i > w(\beta) = \sum \beta_i$ or, $w(\alpha) = w(\beta)$ and $\alpha \succ_{\text{lex}} \beta$.

Definition 23. Let $f(\boldsymbol{X}) = \sum_{\boldsymbol{I}} a_{\boldsymbol{I}} \boldsymbol{X}^{\boldsymbol{I}}$ and $\succ$ a monomial order.

1. The multi-degree of f is $\text{multideg}(f) = \max\{\boldsymbol{I} \mid a_{\boldsymbol{I}} \neq 0\}$ where the maximum is taken w.r.t $\succ$.
2. The leading coefficient of f is $\text{LC}(f) = a_{\text{multideg}(f)} \in \mathbb{F}$.
3. The leading monomial of f is $\text{LM}(f) = \boldsymbol{X}^{\text{multideg}(f)}$.
4. The leading term of f is $\text{LT}(f) = \text{LC}(f) \cdot \text{LM}(f)$.

Definition 24 (Multivariate polynomial division). Let $\succ$ be a monomial order on $\mathbb{Z}^m_{\geq 0}$, and let $F = \{f_1, \ldots, f_k\}$ be a set of k polynomials in $\mathbb{F}[\boldsymbol{X}]$. Then every $f \in \mathbb{F}[\boldsymbol{X}]$ can be written as

$$f = q_1 f_1 + \cdots + q_k f_k + r,$$

where $q_i, r \in \mathbb{F}[\boldsymbol{X}]$, and either $r = 0$ or r is a linear combination, with coefficients in $\mathbb{F}$, of monomials, none of which is divisible by any of $\text{LT}(f_1), \ldots, \text{LT}(f_k)$. We call r a remainder of the division by F. Moreover, $\text{multideg}(q_i f_i) \leq \text{multideg}(f)$ for every $i \in [s]$. The remainder r is not necessarily unique, and might depend on the order of division.

Definition 25. Let $\{0\} \neq I \subseteq \mathbb{F}[\boldsymbol{X}]$ be an ideal. Fix a monomial ordering on $\mathbb{F}[\boldsymbol{X}]$. Then

1. We denote by $\text{LT}(I)$ the set of leading terms of non-zero elements in I.

$$\text{LT}(I) = \{\text{LT}(f) \mid f \in I \setminus \{0\}\}.$$

2. We denote by $\langle \text{LT}(I) \rangle$ the ideal generated by the elements in $\text{LT}(I)$.

Definition 26. Let $\{0\} \neq I \subseteq \mathbb{F}[\boldsymbol{X}]$ be an ideal. Fix a monomial order on $\mathbb{F}[\boldsymbol{X}]$. A subset $G = \{g_1, \ldots, g_t\} \subset I$ is said to be a Grobner basis for I, if

$$\langle \mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_t) \rangle = \langle \mathrm{LT}(I) \rangle.$$

Note 27. Every ideal $I \subset \mathbb{F}[X_1, \ldots, X_m]$ is finitely generated and has a Grobner basis. The importance of a Grobner basis, is that it gives a natural way of choosing representatives for the quotient space $\mathbb{F}[X_1, \ldots, X_m]/I$.

Theorem 28 ([5], Section 2, Proposition 1). Let $I \subset \mathbb{F}[X_1, \ldots, X_m]$ be an ideal and $G = \{g_1, \ldots, g_k\}$ a Grobner basis. Then given $f \in \mathbb{F}[\boldsymbol{X}]$ there is a unique $r \in \mathbb{F}[\boldsymbol{X}]$ such that there is a $g \in I$ such that $f = g + r$ and no term of r is divisible by any of $\mathrm{LT}(g_1), \ldots, \mathrm{LT}(g_k)$. We call this r, the reduced form of f (relative to I). Note that the reduced form of any polynomial is equivalent to this polynomial modulo I. Thus, the theorem gives us a natural way of choosing representatives modulo I.

Theorem 29. Let $R = \mathbb{F}[\boldsymbol{X}]$ be the ring of polynomials, and $I \subset R$ an ideal. Let G be a Grobner basis for I. Then the set

$$B = \{M(\boldsymbol{X}) \mid M \text{ is a monomial not divisible by any } \mathrm{LT}(g) \text{ for } g \in G\},$$

is a basis for R/I. The following criterion determines whether G is a Grobner basis.

Definition 30 (LCM and S polynomials). Let $f, g \in \mathbb{F}[\boldsymbol{X}]$ be non-zero polynomials. Let $\alpha = \mathrm{multideg}(f)$ and $\beta = \mathrm{multideg}(g)$.

1. The least common multiple of $\mathrm{LM}(f)$ and $\mathrm{LM}(g)$, denoted $\mathrm{LCM}(\mathrm{LM}(f), \mathrm{LM}(g))$, is $\boldsymbol{X}^{\gamma}$, where $\gamma = (\gamma_1, \ldots, \gamma_m)$ and $\gamma_i = \max\{\alpha_i, \beta_i\}$ for each i.
2. The S-polynomial of f and g is

$$S(f, g) = \frac{\mathrm{LCM}(\mathrm{LM}(f), \mathrm{LM}(g))}{\mathrm{LT}(f)} \cdot f - \frac{\mathrm{LCM}(\mathrm{LM}(f), \mathrm{LM}(g))}{\mathrm{LT}(g)} \cdot g.$$

Theorem 31 (Buchberger's Criterion, [5], Sec 6). Let $I \subset \mathbb{F}[\boldsymbol{X}]$ be an ideal. Then a basis $G = \{g_1, \ldots, g_k\}$ of I is a Grobner basis of I if and only if for all pairs $i \neq j$, the remainder on division of $S(g_i, g_j)$ by G is zero. Note that we always have $S = S(g_i, g_j) \in I$ by the definition of S. When saying the remainder of the division by G is zero, we mean that there are $\{f_i\}$, such that $S = \sum f_i g_i$ and $\mathrm{multideg}(f_i g_i) \leq \mathrm{multideg}(S)$ for every i (as in Definition 24).

Theorem 32 (Combinatorial Nullstellensatz, [1]). Let $\mathbb{F}$ be a field, and $A_1, \ldots, A_m \subseteq \mathbb{F}$. Let $g_i(X) = \prod_{\alpha \in A_i} (X - \alpha)$ for $i = 1, \ldots, m$. Assume a polynomial $f \in \mathbb{F}[\boldsymbol{X}]$ satisfies $f(\alpha) = 0$ for all $\alpha \in A_1 \times \cdots \times A_m$. Then there are $h_1, \ldots, h_t$ such that

$$f = \sum h_i g_i,$$

and $\deg(h_i) + \deg(g_i) \leq \deg(f)$ for all i. When $A_i = \mathbb{F}_q$ denote

$$g(X) = \prod_{\alpha \in \mathbb{F}_q} (X - \alpha) = X^q - X.$$

Also, let I_m denote the ideal $I_m = \{f \in \mathbb{F}_q[\boldsymbol{X}] \mid \forall \alpha \in \mathbb{F}_q^m \ f(\alpha) = 0\}$.

Corollary 33. $I_m = \langle g(X_1), \ldots, g(X_m) \rangle$.

Proof. Let $f \in I_m$. By Theorem 32, taking $S_i = \mathbb{F}_q$ for every i, we get that $f = \sum h_i g_i$ for some $\{h_i\}$ and so $f \in \langle g_k \rangle_{k \in [m]}$. The other inclusion is trivial since $g(X_i) = X_i^q - X_i$ vanishes on $\mathbb{F}_q^m$ for every i. $\triangleleft$ Using Theorem 31 it can be easily proved that:

Claim 34. $G = \{g(X_k)\}_{k=1}^m$ is a Grobner basis for I_m relative to the total degree lexicographic order. Let $s \in \mathbb{N}$ and let $I_{m,s}$ denote the ideal

$$I_{m,s} = \{f \in \mathbb{F}_q[\boldsymbol{X}] \mid \forall \alpha \in \mathbb{F}_q^m \ \mathrm{Mult}(f; \alpha) \geq s\}.$$

In this notation, $I_{m,1} = I_m$ defined before. For every $\boldsymbol{I} = (I_1, \ldots, I_m) \in \mathbb{N}^m$ define

$$g(\boldsymbol{X})^{\boldsymbol{I}} = \prod_{k=1}^m g(X_k)^{I_k}.$$

Theorem 35 (Combinatorial Nullstellensatz with multiplicity, [3], Sec 3). $I_{m,s} = \langle g(\boldsymbol{X})^{\boldsymbol{I}} \rangle_{w(\boldsymbol{I})=s}$. Furthermore, the set $G_{m,s} = \{g(\boldsymbol{X})^{\boldsymbol{I}}\}_{w(\boldsymbol{I})=s}$ is a Grobner basis for $I_{m,s}$.

Proof. To see that $I_{m,s}$ is indeed an ideal, fix $f \in I_{m,s}$ and $h \in \mathbb{F}[\boldsymbol{X}]$. Then for $r < s$: $(hf)^{(r)} = \sum_{i=0}^r f^{(i)} \cdot h^{(r-i)} = 0$ and so $hf \in I_{m,s}$. Also, clearly, $g(\boldsymbol{X})^{\boldsymbol{I}} \in I_{m,s}$ for every $\boldsymbol{I}$ with $w(\boldsymbol{I}) = s$. We need to show $G_{m,s}$ is a Grobner basis for $I_{m,s}$, [3, Section 3] show $G_{m,s}$ generates $I_{m,s}$ and, furthermore, $f = \sum_{b:w(\boldsymbol{I})=s} g(\boldsymbol{X})^{\boldsymbol{I}} h_{\boldsymbol{I}}$ for some $h_{\boldsymbol{I}}$ with $\deg(h_{\boldsymbol{I}}) \leq \deg(f) - s \deg(g)$. In particular, this is true for the S polynomials in Theorem 31. I.e, every such S polynomial can be expressed as $S = \sum g(\boldsymbol{X})^{\boldsymbol{I}} h_{\boldsymbol{I}}$ where $\deg(g(\boldsymbol{X})^{\boldsymbol{I}} h_{\boldsymbol{I}}) \leq \deg(S)$. By Buchberger's criterion $g(\boldsymbol{X})^{\boldsymbol{I}}$ is a Grobner basis. $\triangleleft$ Finally, we look at equality modulo $I_{m,1}$.

Definition 36. For $n_1, n_2 \in \mathbb{N}$ we say $n_1 =_{\mathbb{F}_q} n_2$ iff $x^{n_1} = x^{n_2} \bmod I_{1,1}$. Equivalently, $n_1 =_{\mathbb{F}_q} n_2$ iff

- $n_1 = n_2$, or,
- $\min(n_1, n_2) > 0$ and $n_1 = n_2 \bmod (q-1)$.

Definition 37. Let $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{N}^m$. We say $\boldsymbol{A} =_{\mathbb{F}_q} \boldsymbol{B}$ for iff $A_k =_{\mathbb{F}_q} B_k$ for every $1 \leq k \leq m$. We also record:

Claim 38. Let $a, b \in \mathbb{N}$. If $a =_{\mathbb{F}_q} b$ and $a < q$ then $a \leq b$.

Proof. If $b < q$ then $a =_{\mathbb{F}_q} b$ implies $a = b$. Otherwise $a < q \leq b$. ◁

3 Polynomials and tables (continued author context)

Note that every polynomial determines its table EVAL(P). However, two different polynomials might have the same table if their difference is the zero table on $\mathbb{F}_q^m$. From Theorem 35,

$$\mathrm{Ker}(\mathrm{EVAL}_{m,s}) = I_{m,s} = \langle g(\boldsymbol{X})^{\boldsymbol{I}} \rangle_{w(\boldsymbol{I})=s}.$$

Since $I_{m,s}$ does not contain any non-zero polynomial of total degree less than sq, we conclude that:

Corollary 40. $\mathrm{EVAL}_{m,s}$ is injective on polynomials of total degree less than sq.

Claim 41. $\mathrm{EVAL}_{m,s}$ is onto $\Sigma_{m,s}^{q^m}$.

Proof. We will show that

$$\dim(\mathbb{F}_q[X_1, \ldots, X_m]/I_{m,s}) \geq \dim((\Sigma_{m,s})^{q^m}) = q^m \cdot \binom{m+s-1}{m},$$

where the dimension is over $\mathbb{F}_q$. This implies that the image of $\mathrm{EVAL}_{m,s}$ is everything, and the mapping is injective. To see that consider the set

$$B_{m,s} = \left\{ g(\boldsymbol{X})^{\boldsymbol{I}} \cdot \prod_{k=1}^{m} X_k^{j_k} \mid w(\boldsymbol{I}) < s, 0 \leq j_k < q \right\}. \tag{5}$$

The elements in $B_{m,s}$ have different multi degree and therefore are independent. Also elements in $B_{m,s}$ are monomials of degree smaller than sq, and therefore are $I_{m,s}$ reduced. Thus,

$$\begin{aligned}
\dim(\mathbb{F}_q[\boldsymbol{X}]/I_{m,s}) &\geq \dim \mathrm{Span}(B_{m,s}) \\
&= |\{(\boldsymbol{I}, \boldsymbol{J}) \in \mathbb{N}^m \times \mathbb{N}^m \mid w(\boldsymbol{I}) < s, 0 \leq j_k < q\}| \\
&= \binom{m+s-1}{m} \cdot q^m,
\end{aligned}$$

as desired. ◁

Corollary 42. The set $B_{m,s}$ is a basis for $\mathbb{F}[\boldsymbol{X}]/I_{m,s}$.

Example 43. When $m = 1$ the set $\{g(x)^i x^j\}_{i \in \mathbb{N}, j<q}$ is a basis of $\mathbb{F}[\boldsymbol{X}]$. Thus, for every $\ell \in \mathbb{N}$ there are $\beta_{\ell,i,j}$ such that

$$x^\ell = \sum_{i_1, i_2 : i_1 q + i_2 \leq \ell, i_2 < q} \beta_{\ell, i_1, i_2} g(x)^{i_1} x^{i_2}.$$

By comparing coefficients of x^ℓ we see that $\beta_{\ell, \lfloor \ell/q \rfloor, \ell \bmod q} = 1$.

Claim 44. For every table $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ there exists a degree $q(s-1)+(q-1)m$ polynomial such that $\mathrm{EVAL}(P) = T$.

Proof. By Claim 41, T may be written as $\mathrm{EVAL}(P)$, where P is an $\mathbb{F}_q$-combination of basis elements in $B_{m,s}$. By the definition of $B_{m,s}$, any basis element is of the form $g(\boldsymbol{X})^{\boldsymbol{I}} \cdot \prod_{k=1}^m X_k^{j_k}$ where $w(\boldsymbol{I}) < s$ and $0 \le j_k < q$. As such, $\deg(P) \le \deg(g(\boldsymbol{X})^{\boldsymbol{I}}) + \deg(\prod_{k=1}^m X_k^{j_k}) \le q(s-1)+(q-1)m$. $\triangleleft$ We will later need:

Lemma 45. Let $A \in \mathbb{F}_q[t]$ and $B \in \mathbb{F}_q[r]$. Then,

$$\deg_t(A(t)B(r) \bmod I_{2,s}) \le \deg_t(A).$$

Proof. Express A in the basis $\{g(t)^i t^j\}_{i\in\mathbb{N}, j<q}$,

$$A(t) = \sum_{(i,j)\in\mathcal{A}} \alpha_{i,j} g(t)^i t^j,$$

and similarly for B,

$$B(r) = \sum_{(k,\ell)\in\mathcal{B}} \beta_{k,\ell} g(r)^k r^\ell,$$

for some sets $\mathcal{A}, \mathcal{B}$, $\alpha_{i,j}, \beta_{k,\ell} \in \mathbb{F}_q$. Then,

$$A(t)B(r) \bmod I_{2,s} = \sum_{(i,j)\in\mathcal{A},(k,\ell)\in\mathcal{B}: i+k<s} \alpha_{i,j}\beta_{k,\ell} g(t)^i g(r)^k t^j r^\ell.$$

Suppose $\deg_t(A(t)B(r) \bmod I_{2,s})$ is achieved on the monomial of $(i,j) \in \mathcal{A}, (k,\ell) \in \mathcal{B}$. In particular, this degree in t is already achieved in A for the monomial (i,j). Thus, $\deg_t(AB) \le \deg_t(A)$. $\triangleleft$ Next, we would like to give a purely algebraic criteria, which states when exactly a table belongs to the code $\mathrm{MRM}(q,m,d,s)$.

Definition 46. Let $T \in \Sigma_{m,s}^{q^m}$ be a table. By Corollary 40 and Claim 41 there is a unique element $P_T \in \mathbb{F}[\boldsymbol{X}]/I_{m,s}$ such that $\mathrm{EVAL}_{m,s}(P_T) = T$. We call P_T the representing polynomial of T.

Lemma 47. Assume $d < sq$. Let $T \in \Sigma_{m,s}^{q^m}$ be a table, and P_T its representing polynomial. Then

$$T \in \mathrm{MRM}(q,m,d,s) \iff \deg(P_T) \le d.$$

Proof. First, assume $\deg(P_T) \le d$. Then, by definition, since $\mathrm{EVAL}_{m,s}(P_T) = T$, we have $T \in \mathrm{MRM}(q,m,d,s)$. For the other direction, assume $T \in \mathrm{MRM}(q,m,d,s)$. Then there is some $Q \in \mathbb{F}[\boldsymbol{X}]$ of total degree $\le d$ such that $\mathrm{EVAL}_{m,s}(Q) = T$. As $\deg(Q) \le d < sq$, Q is $I_{m,s}$ reduced and by Corollary 40 Q is the representing polynomial of T. $\triangleleft$

Imported RM tester and numerical transfer (author context)

Theorem 2 ([12]). Let $k = \lceil (d+1)/(q-q/p) \rceil$. Given $f : \mathbb{F}_q^m \to \mathbb{F}_q$ let:

- $\mathrm{REJ}_{k,d}(f)$ denotes the rejection probability of the test, namely, the probability over a random k dimensional affine space, that f restricted to the space cannot be explained by a degree k polynomial, and,
- $\delta(f, \mathrm{RM}(q,m,d))$ be the distance of f from the code $\mathrm{RM}(q,m,d)$.

Then

$$\mathrm{REJ}_{k,d}(f) \geq \min\{\alpha_0 \cdot \delta(f, \mathrm{RM}(q,m,d)), c_0\}.$$

where $\alpha_0 = q^k/2$ and $c_0 = 1/(2(k+1)q^{k+1})$. Combining the result with Theorem 2 we get:

Corollary 52. Let $d, q, s < q$ be positive integers and let $t = t_{q,d} = \lceil (d+1)/(q-q/p) \rceil$ be the RM testing dimension. Then,

$$\mathrm{REJ}_{t,d}^{\mathrm{MRM}}(T) \geq \min\left\{ \frac{1}{3}\left(1 - \frac{s-1}{q}\right) \delta(T, \mathrm{MRM}(q,m,d,s)), \frac{1}{2(t+1)q^{t+1}} \right\}.$$

Proof. Theorem 2 from [12] ensures that

$$\mathrm{REJ}_{t,d}(f) \geq \min\left\{ \frac{q^t}{2}\delta(f, \mathrm{RM}(q,m,d)), \frac{1}{2(t+1)q^{t+1}} \right\}.$$

Using Theorem 51 we see that

$$\alpha' \geq \left(1 - \frac{s-1}{q}\right) \frac{1}{1 + q^{d/(q-1)}\frac{1}{\alpha}} \geq \frac{1}{3}\left(1 - \frac{s-1}{q}\right).$$

using the fact that $d/(q-1) \leq t$ and $q^{d/(q-1)}/\alpha \leq 2$. ◁ In particular, for $d < (q - q/p)$ the line test is a local characterization, as $t_{q,d} = 1$. We note that the assumption $s \leq d$ is quite natural, as derivatives with order higher than the degree must be identically zero. In contrast, it is not clear whether the assumption $s < q$ is indeed required, and we leave it for future study.

6 The plane test is a characterization

In this section we show that the multiplicity code $\mathrm{MRM}(q,m,d,s)$ can be characterized by restrictions to planes. Let $\mathbb{F}_q$ be a field and let $m, s \leq d$ be positive integers. For $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{F}_q^{m\,1}$ define: $\ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} : \mathbb{F}_q^2 \to \mathbb{F}_q^m$ by:

$$\ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}(t,r) = \boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c}.$$

From Lemma 18 and Definition 48 we see that:

[1] In this section, we omit the requirement that $\boldsymbol{a}, \boldsymbol{b}$ be independent. If some degenerate plane shows that a table is not a low degree polynomial, then some actual plane will too.

Theorem 59 (Completeness). Suppose $d < sq - 1$. If a table $T \in \Sigma_{m,s}^{q^m}$ satisfies $T \in \mathrm{MRM}(q, m, d, s)$ then for all $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{F}_q^m$,

$$\phi_{(\boldsymbol{a},\boldsymbol{b})} \circ T \circ \ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} \in \mathrm{MRM}(q, 2, d, s).$$

The main challenge is proving the converse:

Theorem 60 (Soundness). Suppose q is a power of the prime p, $q \geq s$ and $d < q(s - 1/p)$. If a table $T \in \Sigma_{m,2}^{q^m}$ satisfies that for all $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{F}_q^m$, $\phi_{(\boldsymbol{a},\boldsymbol{b})} \circ T \circ \ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} \in \mathrm{MRM}(q, 2, d, s)$ then $T \in \mathrm{MRM}(q, m, d, s)$. We define the vector space of tables which pass the test:

$$V_{m,d,s} = \{T \in \Sigma_{m,s}^{q^m} \mid \forall \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{F}_q^m \ \phi_{(\boldsymbol{a},\boldsymbol{b})} \circ T \circ \ell_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}} \in \mathrm{MRM}(q, 2, d, s)\}. \quad (12)$$

We denote by $C_{m,d,s} = V_{m,d,s} \setminus \mathrm{MRM}_{m,d,s}$, the set of tables which cheat the test. We would like to show that $C_{m,d,s} = \emptyset$. Assume towards a contradiction that there is a table $T \in C_{m,d,s}$. By Claim 41, T can be realized (uniquely) as an element P of the quotient space $\mathbb{F}_q[\boldsymbol{X}]/I_{m,s}$. We use the basis $B_{m,s}$ from Equation (5) to write P in the form

$$P(\boldsymbol{X}) = \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \cdot g(\boldsymbol{X})^{\boldsymbol{I}} \boldsymbol{X}^{\boldsymbol{J}},$$

where $\alpha_{\boldsymbol{I},\boldsymbol{J}} \in \mathbb{F}_q$ and $(\boldsymbol{I}, \boldsymbol{J}) \in M_{s,q}$ iff $w(\boldsymbol{I}) < s$ and $J_k < q$ for every $1 \leq k \leq m$. Since $T \notin \mathrm{MRM}_{m,d,s}$ we have $\deg(P) > d$. This means there must be some $\boldsymbol{I}$ and $\boldsymbol{J}$ such that $\alpha_{\boldsymbol{I},\boldsymbol{J}} \neq 0$ and

$$w(\boldsymbol{I})q + w(\boldsymbol{J}) > d. \quad (13)$$

We may assume that every $\boldsymbol{I}, \boldsymbol{J}$ for which $\alpha_{\boldsymbol{I},\boldsymbol{J}} \neq 0$ satisfy Equation (13). This is since the test is linear, and any degree $\leq d$ terms have no effect on whether P passes the test or not. We use the following monomial order $\succ_w$ on $B_{m,s}$:

1. First order monomials according to $w(\boldsymbol{I}) + w(\boldsymbol{J})$,
2. Then order monomials according to $w(\boldsymbol{I})$,
3. Finally, order monomials according to the lexicographic order on $\boldsymbol{I}, \boldsymbol{J}$.

We emphasize that $\succ_w$ is not a monomial order in the sense of Grobner bases, and we make no use of it in that sense. Let $(\boldsymbol{I}_{\max}, \boldsymbol{J}_{\max})$ be s.t. $g(\boldsymbol{X})^{\boldsymbol{I}_{\max}} \cdot \boldsymbol{X}^{\boldsymbol{J}_{\max}}$ is a maximal monomial of P according to $\succ_w$. For $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \mathbb{F}^m$ the restriction of P to the plane defined by $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$ is $P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) = P(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c})$. Expressing $P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r)$ in the basis $B_{2,\infty}$:

$$P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) = \sum_{i \in \mathbb{N}, j \in \mathbb{N}, k, \ell < q} A_{i,j,k,\ell}(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \cdot g(t)^i g(r)^j t^k r^\ell,$$

$$P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) \bmod I_{2,s} = \sum_{i+j < s, k, \ell < q} A_{i,j,k,\ell}(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \cdot g(t)^i g(r)^j t^k r^\ell.$$

We view $A_{i,j,k,\ell}(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c})$ as a polynomial in the variables $\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}$.

Lemma 61. For every partition $\boldsymbol{J}_{\max} = \boldsymbol{J}^b_{\max} + \boldsymbol{J}^c_{\max}$ such that:

- $q \cdot w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) \leq qs - 1$, and,
- $\binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}} \neq 0 \bmod p$,

the monomial $\boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}}$ appears with a non-zero coefficient at

$$A_{w(\boldsymbol{I}_{\max}),\lfloor w(\boldsymbol{J}^b_{\max})/q \rfloor, 0, w(\boldsymbol{J}^b_{\max}) \bmod q}(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \bmod I_{3m,1}.$$

Where the ideal $I_{3m,1}$ above is in the variables $a_1, \ldots, a_m, b_1, \ldots, b_m, c_1, \ldots, c_m$.

Proof. We expand $P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r)$. First, $P(\boldsymbol{X}) = \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \cdot g(\boldsymbol{X})^{\boldsymbol{I}} \boldsymbol{X}^{\boldsymbol{J}}$. Thus

$$\begin{aligned}
P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) &= \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \cdot g(\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c})^{\boldsymbol{I}} (\boldsymbol{a}t + \boldsymbol{b}r + \boldsymbol{c})^{\boldsymbol{J}} \\
&= \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \prod_{k=1}^{m} (g(a_k t + b_k r + c_k))^{I_k} \cdot \prod_{i=k}^{m} (a_k t + b_k r + c_k)^{J_k} \\
&= \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \prod_{k=1}^{m} (g(t) a_k + g(r) b_k)^{I_k} \cdot \prod_{i=k}^{m} (a_k t + b_k r + c_k)^{J_k} \\
&= \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \sum_{\substack{\boldsymbol{I}_a + \boldsymbol{I}_b = \boldsymbol{I} \\ \boldsymbol{J}_a + \boldsymbol{J}_b + \boldsymbol{J}_c = \boldsymbol{J}}} \binom{\boldsymbol{I}}{\boldsymbol{I}_a} \binom{\boldsymbol{J}}{\boldsymbol{J}_a, \boldsymbol{J}_b, \boldsymbol{J}_c} \\
&\qquad \boldsymbol{a}^{\boldsymbol{I}_a + \boldsymbol{J}_a} \boldsymbol{b}^{\boldsymbol{I}_b + \boldsymbol{J}_b} \boldsymbol{c}^{\boldsymbol{J}_c} \cdot g(t)^{w(\boldsymbol{I}_a)} g(r)^{w(\boldsymbol{I}_b)} t^{w(\boldsymbol{J}_a)} r^{w(\boldsymbol{J}_b)}. \qquad (14)
\end{aligned}$$

We expand $t^{w(\boldsymbol{J}_a)}$ and $r^{w(\boldsymbol{J}_b)}$ in the basis $B_{1,s}$ as in Example 43 to get:

$$\begin{aligned}
P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(t, r) = \sum_{(\boldsymbol{I},\boldsymbol{J}) \in M_{s,q}} \alpha_{\boldsymbol{I},\boldsymbol{J}} \cdot \sum_{\substack{\boldsymbol{I}_a + \boldsymbol{I}_b = \boldsymbol{I} \\ \boldsymbol{J}_a + \boldsymbol{J}_b + \boldsymbol{J}_c = \boldsymbol{J}}} \binom{\boldsymbol{I}}{\boldsymbol{I}_a} \binom{\boldsymbol{J}}{\boldsymbol{J}_a \ \boldsymbol{J}_b \ \boldsymbol{J}_c} \\
\sum_{\substack{i_1, i_2 \\ i_1 q + i_2 \leq w(\boldsymbol{J}_a) \\ i_2 < q}} \sum_{\substack{j_1, j_2 \\ j_1 q + j_2 \leq w(\boldsymbol{J}_b) \\ j_2 < q}} \beta_{w(\boldsymbol{J}_a), i_1, i_2} \beta_{w(\boldsymbol{J}_b), j_1, j_2} \\
\boldsymbol{a}^{\boldsymbol{I}_a + \boldsymbol{J}_a} \boldsymbol{b}^{\boldsymbol{I}_b + \boldsymbol{J}_b} \boldsymbol{c}^{\boldsymbol{J}_c} \cdot g(t)^{w(\boldsymbol{I}_a) + i_1} t^{i_2} \cdot g(r)^{w(\boldsymbol{I}_b) + j_1} r^{j_2}. \qquad (15)
\end{aligned}$$

We now have a representation in the basis $B_{2,\infty}$. Set:

$$\begin{aligned}
\Delta_{rq} &= \left\lfloor \frac{w(\boldsymbol{J}^b_{\max})}{q} \right\rfloor, \\
\Delta_t &= 0, \\
\Delta_r &= w(\boldsymbol{J}^b_{\max}) \bmod q. \qquad (16)
\end{aligned}$$

We wish to see for which choice of values $\boldsymbol{I}, \boldsymbol{J}, \boldsymbol{I}_a, \boldsymbol{I}_b, \boldsymbol{J}_a, \boldsymbol{J}_b, \boldsymbol{J}_c, i_1, i_2, j_1, j_2$ in Equation (14), $\boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}}$ appears as a coefficient of $A_{(w(\boldsymbol{I}_{\max}), \Delta_{rq}, \Delta_t, \Delta_r)} \bmod$

$I_{3m,1}$. We must have

$$\begin{aligned}
\boldsymbol{I}_{\max} &=_{\mathbb{F}_q} \boldsymbol{I}_a + \boldsymbol{J}_a && \text{By comparing the powers of } \boldsymbol{a}, \text{ remembering } \bmod I_{3m,1},\\
\boldsymbol{J}^b_{\max} &=_{\mathbb{F}_q} \boldsymbol{I}_b + \boldsymbol{J}_b && \text{By comparing the powers of } \boldsymbol{b}, \text{ remembering } \bmod I_{3m,1},\\
\boldsymbol{J}^c_{\max} &=_{\mathbb{F}_q} \boldsymbol{J}_c && \text{By comparing the powers of } \boldsymbol{c}, \text{ remembering } \bmod I_{3m,1},
\end{aligned}$$

where $=_{\mathbb{F}_q}$ was defined in Definition 36. By Claim 38 together with $s < q$ we have

$$w(\boldsymbol{I}_{\max}) \leq w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a), \tag{17}$$
$$w(\boldsymbol{J}^b_{\max}) \leq w(\boldsymbol{I}_b) + w(\boldsymbol{J}_b), \tag{18}$$
$$\boldsymbol{J}^c_{\max} = \boldsymbol{J}_c. \tag{19}$$

It follows that

$$\begin{aligned}
w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}_{\max}) &= w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) + w(\boldsymbol{J}^c_{\max})\\
&\leq w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a) + w(\boldsymbol{I}_b) + w(\boldsymbol{J}_b) + w(\boldsymbol{J}_c)\\
&= w(\boldsymbol{I}) + w(\boldsymbol{J}).
\end{aligned}$$

As $(\boldsymbol{I}_{\max}, \boldsymbol{J}_{\max})$ is maximal it follows that

$$w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}_{\max}) = w(\boldsymbol{I}) + w(\boldsymbol{J}). \tag{20}$$

This, in turn, implies that both inequalities in Equations (17) and (18) are in fact equalities, i.e.,

$$w(\boldsymbol{I}_{\max}) = w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a), \tag{21}$$
$$w(\boldsymbol{J}^b_{\max}) = w(\boldsymbol{I}_b) + w(\boldsymbol{J}_b). \tag{22}$$

We now look at

$$\deg_t \stackrel{\text{def}}{=} \deg_t(g(t\boldsymbol{a})^{\boldsymbol{I}_a} \cdot (t\boldsymbol{a})^{\boldsymbol{J}_a} \cdot g(r\boldsymbol{b})^{\boldsymbol{I}_b} \cdot (r\boldsymbol{b})^{\boldsymbol{J}_b} \cdot \boldsymbol{c}^{\boldsymbol{J}_c} \bmod I_{2,s}).$$

On the one hand, we look for the monomial $\boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}}$ in $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},\Delta_t,\Delta_r}$ mod $I_{3m,1}$, and so we should have $\deg_t = q \cdot w(\boldsymbol{I}_{\max}) + \Delta_t$. On the other hand, by Lemma 45,

$$\begin{aligned}
\deg_t &= \deg_t(g(t\boldsymbol{a})^{\boldsymbol{I}_a}(t\boldsymbol{a})^{\boldsymbol{J}_a} g(r\boldsymbol{b})^{\boldsymbol{I}_b}(r\boldsymbol{b})^{\boldsymbol{J}_b}\boldsymbol{c}^{\boldsymbol{J}_c} \bmod I_{2,s})\\
&\leq \deg_t(g(t\boldsymbol{a})^{\boldsymbol{I}_a}(t\boldsymbol{a})^{\boldsymbol{J}_a})\\
&= q \cdot w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a).
\end{aligned}$$

Thus,

$$q \cdot w(\boldsymbol{I}_{\max}) + \Delta_t \leq q \cdot w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a).$$

Together with Equation (21) we see that $q \cdot w(\boldsymbol{I}_a) + q \cdot w(\boldsymbol{J}_a) + \Delta_t \leq q \cdot w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a)$, and therefore $q \cdot w(\boldsymbol{J}_a) + \Delta_t \leq w(\boldsymbol{J}_a)$ which is possible iff $w(\boldsymbol{J}_a) = 0$ (and $\Delta_t = 0$, which is indeed true, see Equation (16)). Now, $w(\boldsymbol{J}_a) = 0$ implies:

$$w(\boldsymbol{I}_a) = w(\boldsymbol{I}_{\max}), \text{ and,} \qquad \boldsymbol{J}_a = \emptyset.$$

We saw in Equation (20) that $w(\boldsymbol{I}) + w(\boldsymbol{J}) = w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}_{\max})$. If $\boldsymbol{I}_b \neq \emptyset$ then

$$w(\boldsymbol{I}) = w(\boldsymbol{I}_a) + w(\boldsymbol{I}_b) > w(\boldsymbol{I}_a) = w(\boldsymbol{I}_a) + w(\boldsymbol{J}_a) = w(\boldsymbol{I}_{\max}).$$

Thus, $(\boldsymbol{I}, \boldsymbol{J}) \succ_w (\boldsymbol{I}_{\max}, \boldsymbol{J}_{\max})$ in contradiction to the maximality of $(\boldsymbol{I}_{\max}, \boldsymbol{J}_{\max})$. We conclude that $\boldsymbol{I}_b = \emptyset$. As $(\boldsymbol{I}_{\max}, \boldsymbol{J}_{\max}) \in M_{s,q}$ we have $(\boldsymbol{J}_{\max})_k \leq q-1$ for every $k \in [m]$. Thus, $\boldsymbol{J}^b_{\max} =_{\mathbb{F}_q} \boldsymbol{J}_b$, $w(\boldsymbol{J}^b_{\max}) = w(\boldsymbol{J}_b)$ and $\boldsymbol{J}^b_{\max}$ is already reduced. Together this implies that $\boldsymbol{J}^b_{\max} = \boldsymbol{J}_b$. Finally, we use the hypothesis that $q \geq s$. We have, $(\boldsymbol{I}_{\max})_k < s \leq q$ for all $k \in [m]$. Thus, $\boldsymbol{I}_{\max} =_{\mathbb{F}_q} \boldsymbol{I}_a$, $w(\boldsymbol{I}_{\max}) = w(\boldsymbol{I}_a)$ and $\boldsymbol{I}_{\max}$ is already reduced. Together this implies that that $\boldsymbol{I}_{\max} = \boldsymbol{I}_a$. Thus,

$$\boldsymbol{I}_{\max} = \boldsymbol{I}_a = \boldsymbol{I}_a + \boldsymbol{I}_b = \boldsymbol{I}, \qquad \boldsymbol{J}^b_{\max} = \boldsymbol{I}_b + \boldsymbol{J}_b = \boldsymbol{J}_b.$$

Altogether,the only term that may possibly contribute $\boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}}$ to $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},\Delta_t,\Delta_r}$ mod $I_{3m,1}$ is

$$(\boldsymbol{I}_a, \boldsymbol{I}_b, \boldsymbol{J}_a, \boldsymbol{J}_b, \boldsymbol{J}_c) = (\boldsymbol{I}_{\max}, \emptyset, \emptyset, \boldsymbol{J}^b_{\max}, \boldsymbol{J}^c_{\max}).$$

Also, the tuple $(\boldsymbol{I}_a, \boldsymbol{I}_b, \boldsymbol{J}_a, \boldsymbol{J}_b, \boldsymbol{J}_c) = (\boldsymbol{I}_{\max}, \emptyset, \emptyset, \boldsymbol{J}^b_{\max}, \boldsymbol{J}^c_{\max})$ contributes

$$\alpha_{\boldsymbol{I}_{\max},\boldsymbol{J}_{\max}} \cdot \binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}} \sum_{\substack{j_1,j_2 \\ j_1 q + j_2 \leq w(\boldsymbol{J}^b_{\max}) \\ j_2 < q}} \beta_{w(\boldsymbol{J}^b_{\max}),j_1,j_2} \boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}} \cdot g(t)^{w(\boldsymbol{I}_{\max})} \cdot g(r)^{j_1} r^{j_2}$$

to the term in Equation (14). Notice that $w(\boldsymbol{I}_{\max}) + j_1 < s$, for otherwise $q \cdot w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) \geq q(w(\boldsymbol{I}_{\max}) + j_1) \geq qs$ in contradiction to the hypothesis. Thus the term is already $I_{2,s}$ reduced. The contribution to $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},0,\Delta_r}$ mod $I_{3m,1}$ occurs for (j_1, j_2) such that $j_1 = \Delta_{rq}$ and $j_2 = \Delta_r$. Thus, in the sum in Equation (14) there is exactly one possible way to contribute $\boldsymbol{a}^{\boldsymbol{I}_{\max}}\boldsymbol{b}^{\boldsymbol{J}^b_{\max}}\boldsymbol{c}^{\boldsymbol{J}^c_{\max}}$ to $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},0,\Delta_r}$ mod $I_{3m,1}$, and this is when

$$(\boldsymbol{I}_a, \boldsymbol{I}_b, \boldsymbol{J}_a, \boldsymbol{J}_b, \boldsymbol{J}_c) = (\boldsymbol{I}_{\max}, \emptyset, \emptyset, \boldsymbol{J}^b_{\max}, \boldsymbol{J}^c_{\max}), \text{ and}$$
$$(i_1, i_2, j_1, j_2) = (0, 0, \Delta_{rq}, \Delta_r).$$

The coefficient of this term is:

$$\alpha_{\boldsymbol{I}_{\max},\boldsymbol{J}_{\max}} \cdot \binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}} \cdot \beta_{w(\boldsymbol{J}^b_{\max}),\Delta_{rq},\Delta_r}.$$

$\alpha_{\boldsymbol{I}_{\max},\boldsymbol{J}_{\max}} \neq 0$. By assumption $\binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}}$ is non-zero. $\beta_{\ell,\lfloor \ell/q \rfloor,\ell \bmod q} = 1$ (see Example 43) and taking $\ell = w(\boldsymbol{J}^b_{\max})$ shows the coefficient is non-zero. As there is a unique term contributing the monomial with a non-zero coefficient, the monomial cannot cancel in $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},\Delta_t,\Delta_r}$ mod $I_{3m,1}$ and $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},\Delta_t,\Delta_r}$ mod $I_{3m,1}$ is non-zero. ◁

Lemma 62. There is a partition $\boldsymbol{J}_{\max} = \boldsymbol{J}^b_{\max} + \boldsymbol{J}^c_{\max}$, such that

- $d < q \cdot w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) \leq qs - 1$, and,
- $\binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}} \bmod p \neq 0$.

Proof. Suppose $q = p^w$ where p is prime (if q is prime then $p = q$ and $w = 1$). We choose $\boldsymbol{J}^b_{\max}$ as follows. We go over $k = 1, \ldots, m$ and find the first $k_0 \geq 0$ such that $q \cdot w(\boldsymbol{I}_{\max}) + \sum_{k=0}^{k_0} (\boldsymbol{J}_{\max})_k > d$. There must be some $k_0 \leq m$ like that since $qw(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}_{\max}) > d$. We set:

- $(\boldsymbol{J}^b_{\max})_k = (\boldsymbol{J}_{\max})_k$, for $k = 1, \ldots, k_0 - 1$, and,
- $(\boldsymbol{J}^b_{\max})_k = 0$, for $k = k_0 + 1, \ldots, m$.

Set $v = (\boldsymbol{J}_{\max})_{k_0}$. There is a first value $0 < v' \leq v$ such that

$$qw(\boldsymbol{I}_{\max}) + \sum_{k=1}^{k_0-1} (\boldsymbol{J}_{\max})_k + v' > d.$$

We express $v = (\boldsymbol{J}_{\max})_{k_0}$ in base p: $v = \sum_{\ell=0}^{w-1} v_\ell \cdot p^\ell$. We let v'' be the first integer such that:

- $v'' \geq v'$, and,
- If we express v'' in base p as $v'' = \sum_{\ell=0}^{w-1} v''_\ell \cdot p^\ell$ then $v''_\ell \leq v_\ell$ for every ℓ.

Claim 63. $v' \leq v'' \leq \min\{v, v' + p^{w-1} - 1\}$.

Proof. Notice that v respects the conditions that we need, and so if $v \leq v' + p^{w-1} - 1$ the claim holds. Otherwise, $v \geq v' + p^{w-1} - 1$. Then, $v'' \leq v' + p^{w-1} - 1$ because we can increase v' by setting all the lower bits in the p-representation to 0, while increasing the most significant bit (that is multiplied by p^{w-1}) by 1. ◁ Thus, by Lucas theorem (see Equation (4))

$$\binom{v}{v''} \bmod p = \prod_{\ell=0}^{w-1} \binom{v_\ell}{v''_\ell} \neq 0.$$

Having that, we let $(\boldsymbol{J}^b_{\max})_{k_0} = v''$. We have, $\boldsymbol{J}^b_{\max} \leq \boldsymbol{J}_{\max}$, $qw(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) > d$ and $\binom{\boldsymbol{J}_{\max}}{\boldsymbol{J}^b_{\max}} \bmod p \neq 0$. Also,

$$qw(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) \leq d + p^{w-1} \leq qs - 1,$$

because $d \leq qs - p^{w-1} - 1 = q(s - 1/p) - 1$, completing the proof of the lemma. ◁ We are now ready to prove Theorem 60.

Proof. Fix a partition $\boldsymbol{J}_{\max} = \boldsymbol{J}^b_{\max} + \boldsymbol{J}^c_{\max}$ as in Lemma 62. Let

$$\begin{aligned} \deg_r &= \deg_r(g(t)^{w(\boldsymbol{I}_{\max})} r^{w(\boldsymbol{J}^b_{\max})} \bmod I_{2,s}) \\ &= \deg_r(g(t)^{w(\boldsymbol{I}_{\max})} r^{w(\boldsymbol{J}^b_{\max})}) = w(\boldsymbol{J}^b_{\max}). \end{aligned}$$

Define $\Delta_{rq} = \lfloor w(\boldsymbol{J}^b_{\max})/q \rfloor$, $\Delta_t = 0$ and $\Delta_r = w(\boldsymbol{J}^b_{\max}) \bmod q$. By Lemma 61 we know that

$$A_{w(\boldsymbol{I}_{\max}), \Delta_{rq}, \Delta_t, \Delta_r}(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c}) \bmod I_{3m,1} \neq 0.$$

Thus, there exist $\boldsymbol{a}_0, \boldsymbol{b}_0, \boldsymbol{c}_0$ such that

$$A_{w(\boldsymbol{I}_{\max}), \Delta_{rq}, \Delta_t, \Delta_r}(\boldsymbol{a}_0, \boldsymbol{b}_0, \boldsymbol{c}_0) \neq 0.$$

We look at the test $\boldsymbol{a}_0, \boldsymbol{b}_0, \boldsymbol{c}_0$. We have

$$P|_{P_{\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}}}(r,t) \bmod I_{2,s} = \sum_{i+j<s,k,\ell<q} A_{i,j,k,\ell}(\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}) g(t)^i g(r)^j t^k r^\ell.$$

As $A_{w(\boldsymbol{I}_{\max}),\Delta_{rq},\Delta_t,\Delta_r}(\boldsymbol{a}_0,\boldsymbol{b}_0,\boldsymbol{c}_0) \neq 0$ we see that

$$\begin{aligned}\deg(P|_{P_{\boldsymbol{a}_0,\boldsymbol{b}_0,\boldsymbol{c}_0}} \bmod I_{2,s}) &\geq q \cdot w(\boldsymbol{I}_{\max}) + q \cdot \Delta_{rq} + \Delta_t + \Delta_r \\ &= q \cdot w(\boldsymbol{I}_{\max}) + w(\boldsymbol{J}^b_{\max}) > d.\end{aligned}$$

Thus, by Lemma 47, the test $(\boldsymbol{a}_0, \boldsymbol{b}_0, \boldsymbol{c}_0)$ rejects. ◁

7 Planes give a local MRM tester

We restate Theorem 9:

Theorem 64. Suppose q is a prime power, $s \leq q$ and $d < q(s - 1/p)$. Let $T : \mathbb{F}_q^m \to \Sigma_{m,s}$ be a table and let $\delta = \delta(T, \mathrm{MRM}(q,m,d,s))$. Then

$$\mathrm{REJ}^{\mathrm{MRM}}_{2,d}(T) \geq \min\{\alpha\delta, c\}$$

with $\alpha = \Omega(q^{-6s+5})$ and $c = \Omega(q^{-8s+4})$.

Proof. We remind the reader that

$$H_k = \{\boldsymbol{h} = (\boldsymbol{h}_0, \boldsymbol{h}_1, \ldots, \boldsymbol{h}_k) \mid \boldsymbol{h}_0, \ldots, \boldsymbol{h}_k \in \mathbb{F}_q^m, \dim(\mathrm{span}\{\boldsymbol{h}_1, \ldots, \boldsymbol{h}_k\}) = k\}.$$

Let us say that $\boldsymbol{v} \in H_k$ is bad, if the k-dimensional test with $\boldsymbol{v}$ rejects, i.e., $(\phi_{\boldsymbol{v}} \circ T)|_{A_{\boldsymbol{v}}} \notin \mathrm{MRM}(q,k,d,s)$. We let $t = \lceil (d+1)/(q - q/p) \rceil$ be the $\mathrm{RM}(q,m,d,s)$ testing dimension. Selecting a uniform $\boldsymbol{u} \in H_2$ is the same as selecting $\boldsymbol{h} \sim H_t$ and then a uniform $\boldsymbol{u} \in H_2$ such that $A_{\boldsymbol{u}} \subset A_{\boldsymbol{h}}$. Thus,

$$\begin{aligned}\mathrm{REJ}^{\mathrm{MRM}}_{2,d}(T) &= \Pr_{\boldsymbol{u} \sim H_2}(\boldsymbol{u} \text{ is bad}) \\ &\geq \Pr_{\boldsymbol{h} \in H_t}(\boldsymbol{h} \text{ is bad}) \cdot \Pr_{\boldsymbol{u} \in H_2 : A_{\boldsymbol{u}} \subset A_{\boldsymbol{h}}}(\boldsymbol{u} \text{ is bad} \mid \boldsymbol{h} \text{ is bad}).\end{aligned}$$

- By Corollary 52, the probability of picking a bad $\boldsymbol{h} \in H_t$ is:

$$\mathrm{REJ}^{\mathrm{MRM}}_{t,d}(T) \geq \min\left\{\frac{\delta}{3q}, \frac{1}{2(t+1)q^{t+1}}\right\}.$$

- By Theorem 60 we know that for any bad $\boldsymbol{h} \in H_t$ there is at least one $\boldsymbol{u} \in H_2$ such that $A_{\boldsymbol{u}}$ is contained $A_{\boldsymbol{h}}$ and $\boldsymbol{u}$ is bad. Furthermore, if $A_{\boldsymbol{u}} = A_{\boldsymbol{u}'}$ then $\boldsymbol{u}$ is bad iff $\boldsymbol{u}'$ is bad. We look at $A_{\boldsymbol{u}}$. There are $(q^2-1)(q^2-q)q^2$ different $\boldsymbol{u}' \in H_2$ such that $A_{\boldsymbol{u}'} = A_{\boldsymbol{u}}$ (because there are $q^2 - 1$ choices for the first basis element, $q^2 - q$ choices for the second basis element and $|A_{\boldsymbol{u}}| = q^2$ choices for the offset).

Altogether,

$$\mathrm{REJ}^{\mathrm{MRM}}_{2,d}(T) \geq \frac{\Omega(q^6)}{q^{3t}} \cdot \min\left\{\frac{\delta}{3q}, \frac{1}{2(t+1)q^{t+1}}\right\} = \Omega\left(\min\left\{q^{-3t+5}\delta, \frac{q^{-4t+5}}{t}\right\}\right).$$

We notice that $t \leq \lceil q(s-1/p)/(q-q/p) \rceil \leq \lceil s/(1-1/p) \rceil \leq 2s$. Thus $\alpha = \Omega(q^{-6s+5})$ and $c = \Omega(q^{-8s+4})$ as promised. ◁

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